CS1060 HW2 Design Document

Collaborators

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The Idea

My idea is a ride-sharing platform that connects people in the same neighborhood or community who are traveling in similar directions. Unlike Uber or Lyft, which rely on professional gig drivers, our system enables true carpooling: riders share trips in privately owned vehicles, splitting costs instead of paying fares. The platform uses AI to optimize matching, ensuring minimal detours, fair cost-splitting, and enhanced convenience.

Existing Services

Uber/Lyft: Primarily commercial ridesharing with drivers who earn income, not cost-sharing between peers. Carpool Apps (e.g., BlaBlaCar): More common for intercity travel, less for daily commuting.
Traditional group-chat-based Carpools: Informal, limited to prearranged groups (coworkers, neighbors).

Differentiation

Our service emphasizes community-based daily commuting: True cost-sharing, not profit-driven. Dynamic AI matching of neighbors heading in the same direction at similar times. Incentives for regular carpools to build trust and reliability. Environmental and cost benefits over single-occupancy rides or professional ride-hailing.

Stakeholders

- **Customers (riders/passengers):** Riders value affordability, convenience, and sustainability. They are motivated by reducing commuting costs compared to driving alone, taxis, or Uber/Lyft, while also saving time through efficient ride-matching. They might also be motivated by environmental concerns, choosing ride-sharing to reduce their carbon footprint. Expected behaviors include frequent use of the service during commute hours, leaving comments for drivers, etc.
- **Users (drivers offering rides):** Drivers value cost-sharing and community connection. They are motivated by lowering their own fuel and parking expenses, reducing the stress of solo driving, and occasionally meeting new people. Some may also see it as a way to support their neighbors or contribute to sustainability.

Expected behaviors include offering regular rides for predictable schedules, adhering to safe driving standards, and maintaining courteous interactions with passengers.

- **Regulators (city transportation agencies, safety authorities):** Regulators value public safety, traffic reduction, and emissions control. They are motivated by ensuring ride-sharing services align with citywide goals like reducing congestion, supporting public transit, and protecting rider safety. Expected behaviors include requiring driver and passenger verification, mandating insurance coverage, enforcing accessibility requirements, etc.
- **Investors (impact-focused funds, civic tech sponsors):** Investors value scalability, innovation, and social responsibility. They are motivated by funding solutions that demonstrate both profitability and measurable social or environmental impact. Expected behaviors include pushing for steady growth, encouraging adoption through partnerships, etc.
- **Owners (startup founders):** Owners value trust, adoption, and long-term sustainability of the business model. They are motivated by establishing a strong reputation, achieving widespread user growth, and meeting regulatory standards. Expected behaviors include securing funding, ensuring safety features are prioritized, and negotiating with regulators and transit agencies.
- **Employees (engineers, support staff, community managers):** These employees value career growth, fair compensation, and meaningful work. They are motivated by the chance to shape innovative mobility technology, build strong communities, and improve urban life. Expected behaviors include continuously improving the app experience, supporting riders and drivers through customer service, etc.

User Journey

Daily Commuter Profile: Aaron, a 25-year-old student at Brown, drives over 50 miles to attend classes at Harvard every Tuesday and Thursday.

- **Planning**: On Monday night, Aaron posts his Tuesday morning trip on the ride-sharing app from Providence to Cambridge. He sets a flat price of $12 per rider, significantly cheaper than an Uber or Lyft for the same distance (over $60).
- **Matching**: By morning, two riders have accepted his offer — both are young professionals commuting to Boston. The app shows Aaron’s total expected earnings ($24), enough to cover most of his fuel costs for the trip.
- **Pickup**: Aaron drives to a designated meeting spot near an I-95 park-and-ride lot. His passengers meet him there, keeping detours minimal. The app verifies check-ins and provides navigation for the shared route.

- **Ride**: The group chats during the drive, and Aaron appreciates that payments are handled automatically through the app. Riders are happy knowing they're paying far less than Uber while still getting a reliable door-to-door commute.
- **Arrival**: Aaron drops his passengers at Kendall Square before continuing to Harvard. He earned money that covers his gas and tolls while maintaining a flexible schedule. Riders saved significant money compared to alternatives, creating a win-win exchange.

Values Tensions

Aaron values earning extra income while offsetting his commute expenses, but passengers value low cost, which could create tension if demand pushes prices up. Riders want safety and reliability, but regulators may step in to ensure Aaron is properly licensed and insured, potentially adding costs or complexity for him. Investors may push to increase monetization (higher commissions or surge pricing), but Aaron and his passengers want the service to remain affordable and community-focused.